Extended Essay

Title: "The Goat Problem"

RQ: "How to represent the surface area, inside a circular fence, to which a goat attached to a rope has access?"

Subject: Mathematics AA HL

Word Count: 3050

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1. Introduction

1.1. The Situation

As mentioned in the title, the situation being analyzed in this work consists of a goat enclosed by a circular fence. Furthermore, the goat is attached to the fence by a rope:

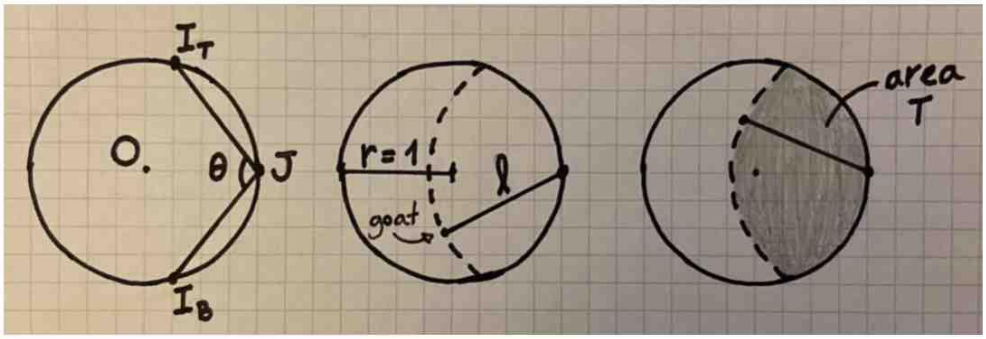


Figure 1: three representations of the circular fence, yet, each time with different features.

By visualizing the situation, it is important to notice that:

- The radius of the circle (formed by the fence) is equal to one: $r = 1$;
- The length of the rope is represented by: l;
    - When totally stretched: $2 \geq l > 0$
- The surface area which the goat has access to is represented by: T;
- The point, in the fence, which the rope is attached to is represented by: J;
- The point where the goat touches the fence when the rope is totally stretched is represented by: I;
    - Since a circle is symmetrical, there will be two points "I"; one at the top and one at the bottom: I_T and I_B, respectively.
    - Observation: in various moments throughout this work, only the top part of the circular fence will be analyzed. For those cases, I_T will be represented simply by I.

- The angle θ is formed by lines $\overline{I_T J}$ and $\overline{I_B J}$. This angle naturally depends on the length of the rope (when totally stretched).
- The center of the circle is represented by: O.

1.2. The Goal

The Goat Problem is, essentially, a geometry problem that looks quite simple at first moment. The original problem, which this work is based on, is a little bit different. In that case, the goal was to find the length of the rope to which the surface area (to which the goat has access) is equal to half of the area of the circular fence (which is: $\frac{\pi}{2}$ squared units).

An even older problem, which is similar, is the: Exterior Grazing Problem. It appeared, for the first time, in the 1748 edition of the English annual journal The Ladies' Diary: or, the Woman's Almanack[1]. The goal was to find the surface area of which a horse attached to a circular fence had access to. However, in that case, the horse was outside of the circular fence and couldn't get in.

In contrast, the purpose of this work is to find functions which represent the total surface area which the goat has access to, shown in *Figure 1* (from this point on, this surface area will be referred to as "*area T*" or just simply "*T*"). The idea is to have T in terms of θ; in other words: $T(\theta)$.

In the beginning, some essential mathematical concepts will be presented. Then, geometric relations inside the circular fence will be identified. By doing this process, at some point, there will be enough information to finally establish a formula. This way, in the first moment, a trigonometric approach to the problem will be shown and, therefore, the first function will have been found. Nevertheless, this is not the only way to represent *area T*.

[1] "Goat grazing problem - Wikipedia." https://en.wikipedia.org/wiki/Goat_grazing_problem. Accessed 13 Jan. 2023.

A second function for T can be found by applying Maclaurin series into the first function. This simply means that the first function, which is a trigonometric function, will be approximated into a polynomial, for the desired interval. At this point, two representations for *area T* will have been found. Nevertheless, only one approach was shown.

Lastly, a third function can be found by using algebra and calculus. In order to find it, the situation has to be approached differently. In this case, the circular fence will be represented in a Cartesian plane. Then, algebraic and calculus methods will be applied to find the third function for *area T*.

1.3. The Relevance Of the Topic

The problem being addressed in this work is rather simple to understand and not really complicated to solve. Nevertheless, it is a very complete problem. It can be approached in different ways, using different areas in mathematics. Thus, a good exploration can be carried out from it.

Trying to get the same result through different methods/ways is an important idea present in the production of knowledge: confirmation of knowledge, to be more precise. In order to check if the piece of information produced is consistent and valid, one may try to find it through a different approach.

Take into consideration Einstein and his Theory of General Relativity. In the first moment, the German scientist used mathematics to formulate the theory. To see if he was correct, he approached the situation differently: he watched an eclipse. In the end, the eclipse seen matched his mathematical predictions, meaning he had in fact produced knowledge. By approaching the situation mathematically and empirically, Einstein was able to confirm his theory.

In this work, the situation is similar. Firstly, as previously mentioned, a representation for *area T* will be found through a geometric approach. Then, in order to confirm this first representation, the problem will be solved with once again, however, with Algebra and Calculus this time.

1.4. Basic Principles, Concepts, and Formulas

In this section, the basic mathematics needed to understand this work will be presented. The formulas here will be stated, however not explained or derived, since they're content within the IB course and are not the focus of the investigation.

Geometry:

- Surface areas (triangle, circle, section of circle)
  - $S_{triangle} = \frac{bh}{2}$ or $S_{triangle} = \frac{ab \cdot sin(\widehat{C})}{2}$
  - $S_{circle} = \pi r^2$
  - $S_{section} = \frac{\theta}{2} r^2$

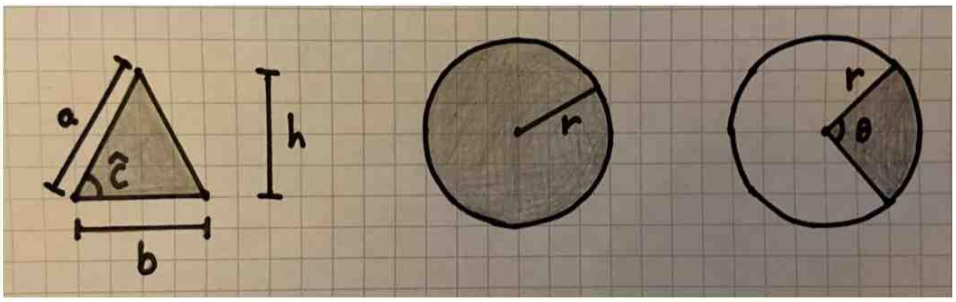


Figure 2: surface areas of (from left to right) the triangle, circle and section of a circle shaded.

- Pythagorean identity
  - $(sin\,\theta)^2 + (cos\,\theta)^2 = 1$
- Cosine rule

- $a^2 = b^2 + c^2 - 2bc(cos\,\widehat{A})$

- Sine rule
  - $\frac{a}{sin\,\widehat{A}} = \frac{b}{sin\,\widehat{B}} = \frac{c}{sin\,\widehat{C}}$

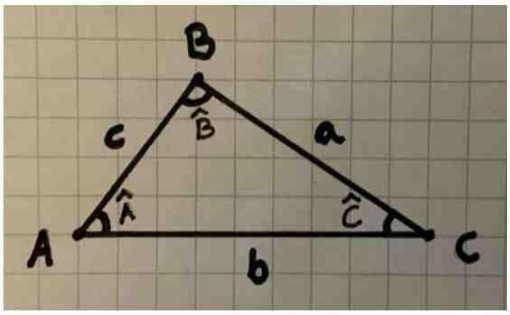


Figure 3: triangle ABC to visualize sine and cosine rules.

- Double Angle Identity
  - For sine → $sin(2\theta) = 2sin(\theta)cos(\theta)$
  - For cosine → $cos(2\theta) = 2cos^2(\theta) - 1$
- Pythagorean Theorem → In right-angled triangles, the square of the opposite side to the right-angle is equal to the sum of the squares of the two other sides:
  - $a^2 = b^2 + c^2$

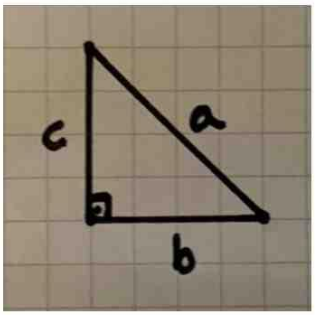


Figure 4: right triangle to visualize the Pythagorean Theorem.

- Length of an arc

- $l = r\theta$

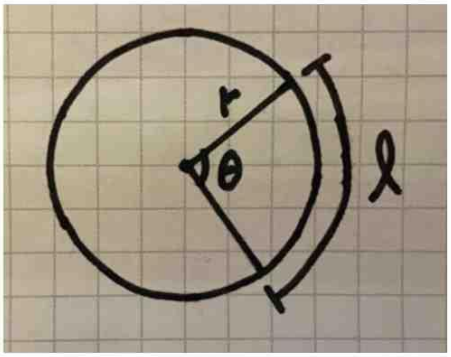


Figure 5: representation of an arc.

Calculus:

- Power rule
  - $f(x) = x^n \rightarrow f'(x) = nx^{n-1}$
- Chain rule
  - $(f \circ g)'(x) = f'(g(x)) \times g'(x)$
- Product rule
  - $y = uv \rightarrow \frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$
- Maclaurin (Taylor series)
  - $f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f^3(0) + (\ldots)$

Algebra:

- Equation of a circle
  - $x^2 + y^2 = r^2$
    - x, y → coordinates of cartesian plane
    - r → radius of the circle

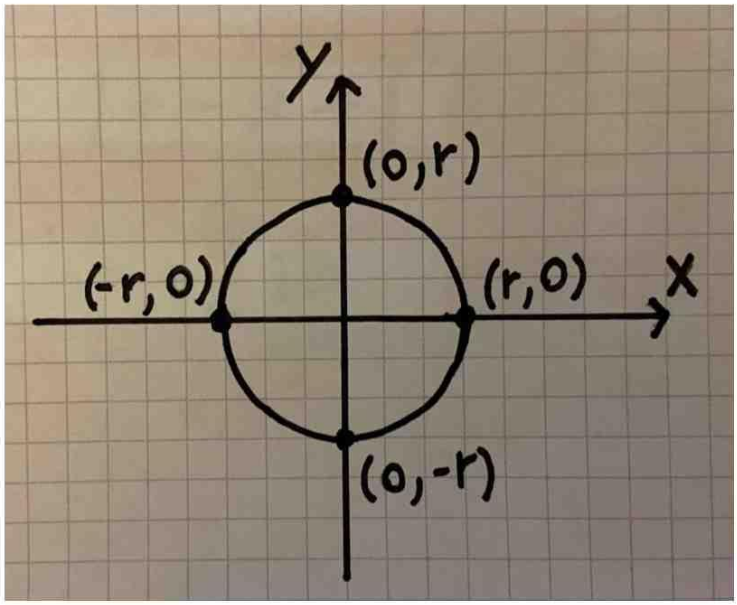


Figure 6: equation of a circle plotted on a Cartesian plane. The points and coordinates of the y- and x-intercepts are also shown.

2. Trigonometric Approach

2.1. Discussing Intervals

For the situation presented in topic 1.1, it is important to have a notion about some trigonometric relations and intervals before going any further into the problem. To start, two new imaginary lines will be formed:

1) $\overline{OI_T}$ → connects the center of the circular fence O to point I in the top part;

2) $\overline{OI_B}$ → connects the center of the circular fence O to point I in the bottom part.

The angle between these two new lines ($\overline{OI_T}$ and $\overline{OI_B}$) will be represented by: λ.

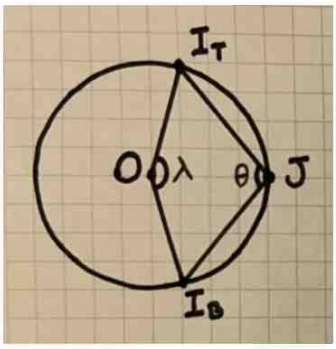


Figure 7: representation of the circular fence, with the two new lines.

By analyzing the image above, it is clear that the angles θ and λ are related. However, their relationship will be only further explored in topic 2.4. For now, the intervals of these two angles will be established:

- $0 \leq \theta < \pi$
- $2\pi \geq \lambda > 0$

Cases:

1) If $\theta = 0$, then $\lambda = 2\pi$ and $l = 2$ (which is the max length). Furthermore, there would be only one point for which the goat touches the fence when the rope is fully stretched, thus $I_T = I_B$. In this case, *area T* would be maximized since the goat would have access to every inch inside the fence.
2) If $\theta \geq \pi$, then the rope would go through the fence and be outside the circular enclosure, which cannot happen as the fence is there to prevent the goat from going outside of the delimited space. In this case, there wouldn't be points I_T and I_B, and the length of the rope l wouldn't depend on θ anymore. Lastly, if $\theta = \pi$, lines $\overline{I_T J}$ and $\overline{I_B J}$ would form a tangent to the circular fence at point J.
3) As λ approaches 0, θ and l will also approach 0. This way, $I_T = I_B$ at J, which means the goat would be fixed at point J. Therefore, there wouldn't be a surface area T since this would be an adimensional situation.

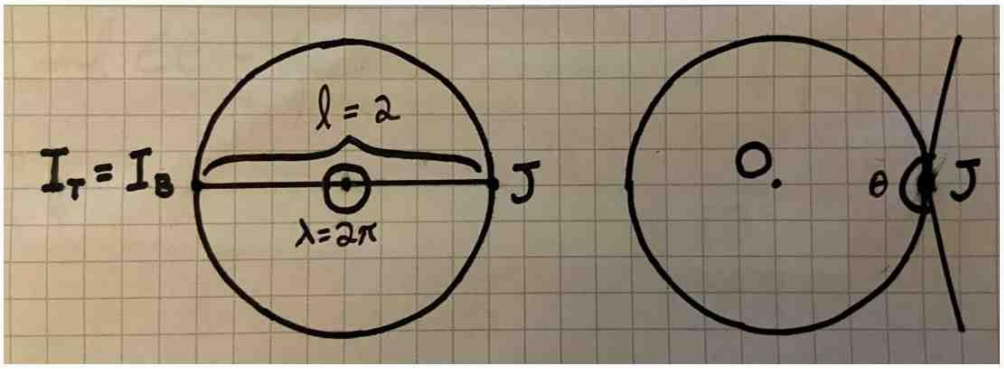


Figure 8: case 1) on the left and case 2) on the right.

Now, a new line will be formed:

- $\overline{OJ}$ → connects the center of the circular fence to the point where the rope is attached in the fence;

Hence, considering only the top part of the circle, a new triangle will be formed → OIJ:

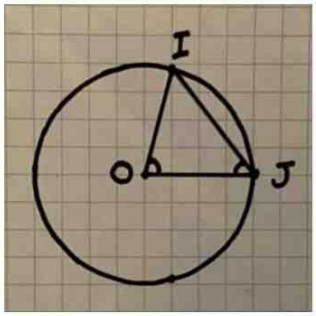


Figure 9: (imaginary) triangle OIJ inside the circular fence.

Since $\overline{OJ}$ and $\overline{OI}$ have their origin at the center of the circular fence and end up touching the arc of the circle, both are radii:

$$\overline{OJ} = \overline{OI} = r = 1$$

Furthermore, $\overline{OJ}$ divides θ and λ in the middle. Thus, for OIJ:

$$\widehat{OJI} = \frac{\theta}{2} \quad \text{and} \quad \widehat{IOJ} = \frac{\lambda}{2}$$

So, the new triangle can be redrawn as:

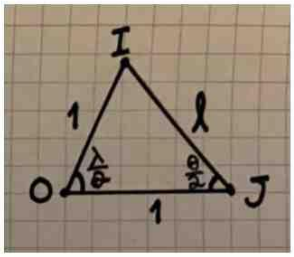


Figure 10: representation of triangle OIJ with two of its inner angles.

Cases:

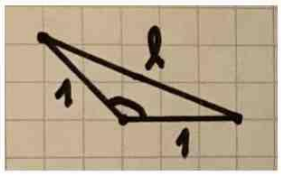
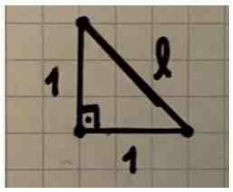
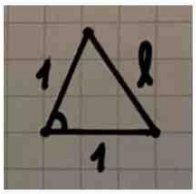
$\lambda > \pi$	$\lambda = \pi$	$\lambda < \pi$
$\theta < \frac{\pi}{2}$	$\theta = \frac{\pi}{2}$	$\theta > \frac{\pi}{2}$
Obtuse	Right	Acute
		

Table 1: cases of the triangle OIJ depending on its inner angles θ *and* λ.

Throughout this work, triangle OIJ will always be represented as an acute triangle, for practical reasons. However, it's important to be aware that OIJ can be any of the three types (obtuse, right and acute), as presented on the table above; it only depends on the length of the rope l (and, consequently, on θ).

2.2. Relating the length of the rope to the angle

In order to find future trigonometric relations inside the circular fence, it is of great importance to know the relation between the length of the rope l and the angle θ. To start, consider OIJ (presented in topic 2.1):

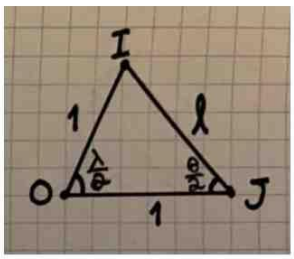


Figure 11: triangle OIJ once again.

Cosine rule can be applied to relate the unknown side to an angle:

$$1^2 = 1^2 + l^2 - 2 \cdot 1 \cdot l \cdot cos(\frac{\theta}{2})$$

$$1 = 1 + l^2 - 2l \cdot cos(\frac{\theta}{2})$$

$$l^2 = 2l \cdot cos(\frac{\theta}{2})$$

$$l = 2 \cdot cos(\frac{\theta}{2})$$

$$l = 2cos(\frac{\theta}{2})$$

From this point on, the relation found above will be referred to as *relation 1*.

2.3. Breaking Down the Areas

Before identifying the other trigonometric relations, *area T* will be broken down into smaller surface areas, in order to simplify the problem:

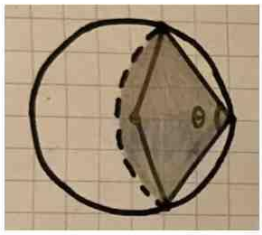
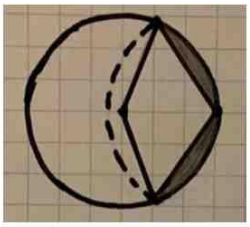
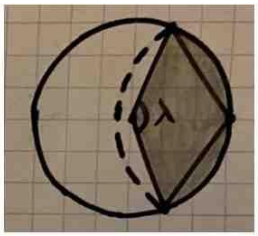
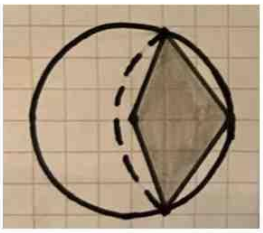
A	B	C	D
			

Table 2: each surface area is represented by the shaded region on the figures.

By analyzing the surface areas above, it is quite clear that T was split into A and B:

$$T = A + B$$

The relation above will be referred to as *relation 2*.

Furthermore, areas C and D are also important, because they will help finding B, since:

$$B = C - D$$

The relation above will be referred to as *relation 3*.

2.4. Finding the Surface Areas

Firstly, the surface area A will be established. By simply observing the figures in *Table 3*, it is possible to see that area A is nothing but a section of a circle with radius l. Thus:

$$A = \frac{\theta}{2} \cdot (l)^2$$

Using *relation 1*:

$$A = \frac{\theta}{2} \cdot (2cos(\frac{\theta}{2}))^2$$

$$A = \frac{\theta}{2} \cdot 4(cos(\frac{\theta}{2}))^2$$

$$A = 2\theta(cos(\frac{\theta}{2}))^2$$

Applying the Double Angle Identity for cosine:

$$A = 2\theta(\frac{cos(\theta)+1}{2})$$

$$A = \theta(cos(\theta) + 1)$$

Since the surface area A was found in terms of θ, it can be re-written as:

$$A(\theta) = \theta(cos(\theta) + 1)$$

Then, there is the surface area B. It is quite difficult, if possible, to find B directly. Thereby, surface areas C and D will be found beforehand. Starting with C:

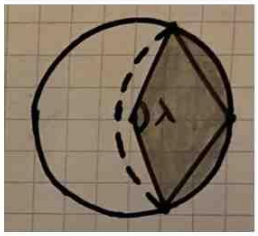


Figure 12: representation of surface area C (shaded region).

For this surface area, finding a relation between θ and λ will be necessary. Considering only the top part of the circular fence, triangle *OIJ* can, once again, be analyzed.

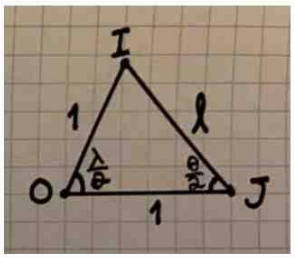


Figure 13: triangle OIJ once again.

Applying cosine rule:

$$l^2 = 1^2 + 1^2 - 2 \cdot 1 \cdot 1 \cdot cos(\frac{\lambda}{2})$$

$$l^2 = 2 - 2cos(\frac{\lambda}{2})$$

Using *relation 1*:

$$(2cos(\frac{\theta}{2}))^2 = 2 - 2cos(\frac{\lambda}{2})$$

$$4(cos(\frac{\theta}{2}))^2 = 2 - 2cos(\frac{\lambda}{2})$$

$$2(cos(\frac{\theta}{2}))^2 = 1 - cos(\frac{\lambda}{2})$$

Applying the Double Angle Identity for cosine:

$$2(\frac{cos(\theta)+1}{2}) = 1 - cos(\frac{\lambda}{2})$$

$$cos(\theta) + 1 = 1 - cos(\frac{\lambda}{2})$$

$$cos(\theta) = -cos(\frac{\lambda}{2})$$

$$cos(\theta) = cos(\pi - \frac{\lambda}{2})$$

$$\theta = \pi - \frac{\lambda}{2}$$

$$\frac{\lambda}{2} = \pi - \theta$$

The relation above will be referred to as *relation 4*.

Now, as the relation between λ and θ has been established, it is possible to find surface area C in terms of θ as well. Just like the surface area A, C is a section of a circle, however with a radius equal to one. Thus:

$$C = \frac{\lambda}{2}(1)^2$$

$$C = \frac{\lambda}{2}$$

Using *relation 4*:

$$C = \pi - \theta$$

$$C(\theta) = \pi - \theta$$

With C established, D will be found by using triangle OIJ as well:

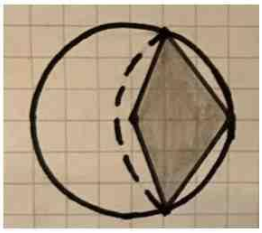


Figure 14: representation of surface area D (shaded region).

By analyzing the figure above, it is clear that the surface area of OIJ (S_{OIJ}) is half of the surface area D:

$$D = 2(S_{OIJ})$$

This way, the surface area of OIJ needs to be found. Using the formula for the surface area of a triangle (presented in topic 1.4):

$$S_{OIJ} = \frac{1}{2} l \cdot 1 \cdot sin(\frac{\theta}{2})$$

$$S_{OIJ} = \frac{l}{2} sin(\frac{\theta}{2})$$

Thus, D can be rewritten as:

$$D = 2(S_{OIJ})$$

$$D = 2(\frac{l}{2} sin(\frac{\theta}{2}))$$

$$D = l \cdot sin(\frac{\theta}{2})$$

Using *relation 1*:

$$D = (2cos(\frac{\theta}{2})) \cdot sin(\frac{\theta}{2})$$

$$D = 2sin(\frac{\theta}{2})cos(\frac{\theta}{2})$$

Applying the Double Angle Identity for sine:

$$D = sin(2 \cdot \frac{\theta}{2})$$

$$D = sin(\theta)$$

$$D(\theta) = sin(\theta)$$

And now, since C and D are established, the surface area B can finally be found. Using *relation 3*:

$$B = C - D$$

$$B = (\pi - \theta) - (sin(\theta))$$

$$B(\theta) = \pi - \theta - sin(\theta)$$

2.5. Determining the 1st Function

Now, there is finally enough information to find T in terms of θ. Using *relation 2*:

$$T = A + B$$

$$T = (\theta(cos(\theta) + 1)) + (\pi - \theta - sin(\theta))$$

$$T = \theta cos(\theta) + \theta + \pi - \theta - sin(\theta)$$

$$T(\theta) = \theta cos(\theta) + \pi - sin(\theta)$$

Plotting this function on a Cartesian plane, for $\{0 \leq \theta < \pi\}$:

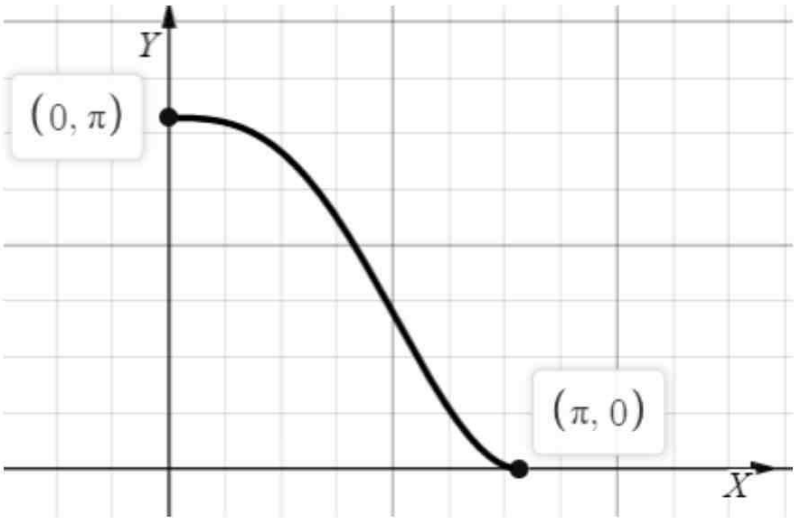


Figure 15: 1st function for area T plotted on a Cartesian plane.

2.6. Finding the polynomial (2nd Function)

The first function has been found. However, it is a transcendental function, since it contains θ and $sin(\theta)$ at the same time. So, by using differentiation techniques and Maclaurin series, the 1st function will be represented as a polynomial. Inevitably, this will be a second representation of *area T*.

In the first place, the 1st function ($T(\theta)$) needs to be found at $x = 0$:

$$T(0) = 0 \cdot (cos(0)) + \pi - sin(0)$$

$$T(0) = 0 + \pi - 0$$

$$T(0) = \pi$$

Then, the derivatives ($T^{n}(\theta)$ for $n \in \mathbb{N}$) need to be found. Applying the power rule and the product rule:

$$T'(\theta) = cos(\theta) + \theta(-sin(\theta)) + 0 - cos(\theta)$$

$$T'(\theta) = -\theta sin(\theta)$$

The derivative also needs to be found at $x = 0$:

$$T'(0) = -0 \cdot sin(0)$$

$$T'(0) = 0$$

By repeating this process, the further derivatives ($n = 2, 3, 4, (...)$) and their value at $x = 0$ can be found as well. The amount of derivatives necessary depends on how precise the polynomial is at the wanted interval ($0 \leq \theta < \pi$); in this case, it will be necessary to find up to the 11th derivative ($T^{11}(\theta)$):

Expressions		Value at $x = 0$
$T(\theta)$	$\theta(cos(\theta)) + \pi - sin(\theta)$	π
$T'(\theta)$	$-\theta sin(\theta)$	0
$T''(\theta)$	$-sin(\theta) - \theta cos(\theta)$	0
$T^{3}(\theta)$	$\theta sin(\theta) - 2cos(\theta)$	-2
$T^{4}(\theta)$	$\theta cos(\theta) + 3sin(\theta)$	0
$T^{5}(\theta)$	$4cos(\theta) - \theta sin(\theta)$	4
$T^{6}(\theta)$	$-5sin(\theta) - \theta cos(\theta)$	0

$T^7(\theta)$	$\theta sin(\theta) - 6cos(\theta)$	-6
$T^8(\theta)$	$\theta cos(\theta) + 7sin(\theta)$	0
$T^9(\theta)$	$8cos(\theta) - \theta sin(\theta)$	8
$T^{10}(\theta)$	$-9sin(\theta) - \theta cos(\theta)$	0
$T^{11}(\theta)$	$\theta sin(\theta) - 10cos(\theta)$	-10

Table 3: values, at $x = 0$, of $T(\theta)$ and its derivatives (up to the 11th derivative).

Now, these values can be input in the Maclaurin series:

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f^3(0) + (\dots)$$

$$T(\theta) = T(0) + \theta T'(0) + \frac{\theta^2}{2!}T''(0) + \frac{\theta^3}{3!}T^3(0) + (\dots)$$

$$T(\theta) = \pi + \theta \cdot 0 + \frac{\theta^2}{2} \cdot 0 + \frac{\theta^3}{6} \cdot (-2) + (\dots)$$

$$T(\theta) = \pi - \frac{1}{3}\theta^3 + \frac{1}{30}\theta^5 - \frac{1}{840}\theta^7 + \frac{1}{45360}\theta^9$$

Plotting this function on a Cartesian plane, for $\{0 \leq \theta < \pi\}$:

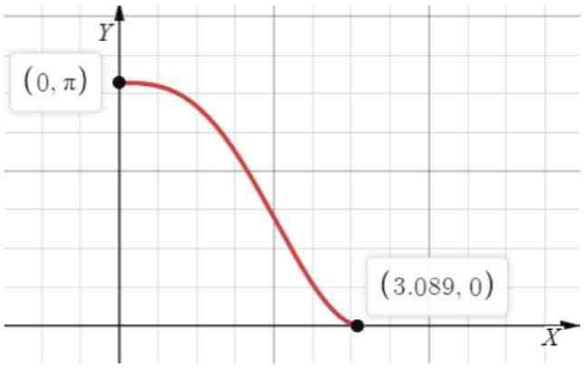


Figure 16: 2nd function for area T plotted on a Cartesian plane.

Comparing to *Figure 15*, it is possible to see that this polynomial representing *area T* has satisfactory precision. Yet, since it is an approximation, it is not perfect, and a clear difference can be seen, which gets bigger and bigger when approaching the ends of the curves:



Figure 17: plotting 1st (in black) and 2nd (in red) equations on the same axes. Zoom around $x = \pi$.

3. Algebraic/Calculus Approach

3.1. Reframing the Situation algebraically

After using trigonometry, the same solution can be found through a different path. Now, calculus and algebra will be used. Firstly, the situation needs to be understood algebraically. So, the circular fence will be plotted in a Cartesian plane, by using the equation of a circle:

$$x^2 + y^2 = r^2$$

As $r = 1$:

$$x^2 + y^2 = 1$$

Yet, in order to apply calculus in the future, a function is preferred. This way, the equation of the circular fence can be rearranged into a function, where y is in terms of x:

$$y^2 = 1 - x^2$$

$$y = \pm\sqrt{1 - x^2}$$

Here, it is important to notice that the top part of the circle is represented by $y > 0$, while the bottom part is represented by $y < 0$.

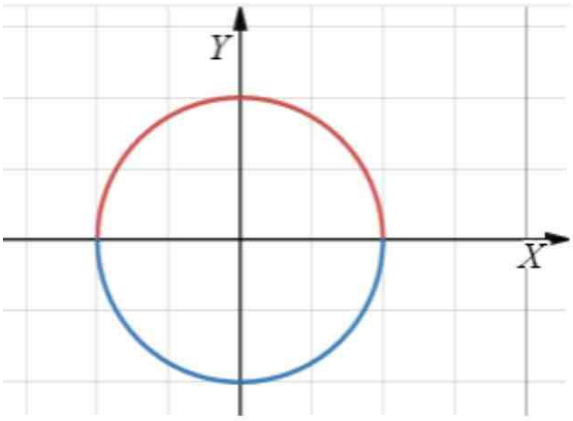


Figure 18: $y > 0$ in red and $y < 0$ in blue.

Lastly, this function of the circular fence will be represented by "p":

$$p(x) = \pm\sqrt{1 - x^2}$$

This way, we end up with a graph like this:

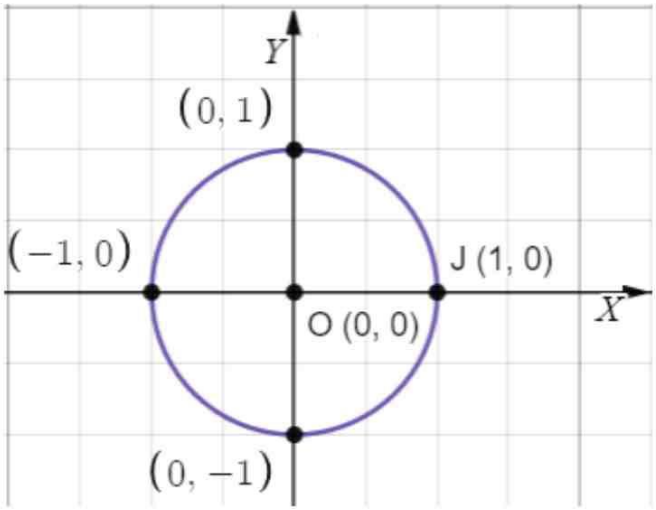


Figure 19: equation for the circular fence plotted on a Cartesian plane.

Analyzing this function and by observing the top part of the graph, point J can be defined:

- $J \rightarrow (1,\ 0)$

On the other hand, the rope of the goat needs to be represented as well. As previously seen with area A, when the rope (fully stretched) travels from I_T to I_B, it creates a section of a circle.

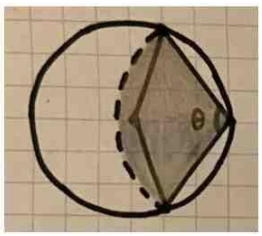


Figure 20: representation of surface area A (shaded region).

This way, to represent the rope and its path (when fully stretched), another equation of a circle will be used. However, in this case, $r = l$:

$$x^2 + y^2 = r^2$$

$$x^2 + y^2 = l^2$$

Furthermore, the center of this circle will be where the rope is attached to the fence → point J. Hence, the equation needs to translate one unit to the positive-end of the x-axis:

$$(x - 1)^2 + y^2 = l^2$$

By doing the same process as before, this equation can also be transformed into a function:

$$y = \pm\sqrt{l^2 - (x - 1)^2}$$

In this case, the function for the rope will be referred to as "q":

$$q(x) = \pm\sqrt{l^2 - (x - 1)^2}$$

Plotting $p(x)$ and $q(x)$ (with an random value for l, as an example) on the same axes:

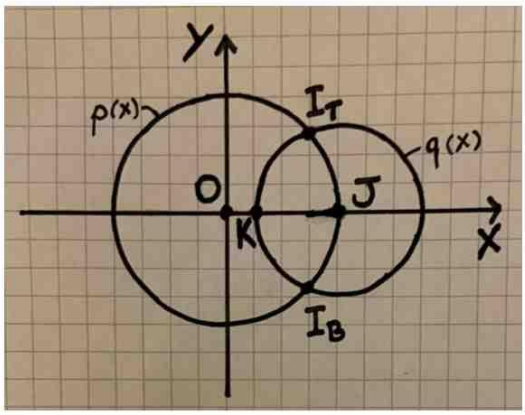


Figure 21: $p(x)$ and $q(x)$ plotted on the same axes, with their intersection points.

The points of intersection between $p(x)$ and $q(x)$ are I_T and I_B. Furthermore, the point where $p(x)$ intercepts the x-axis (inside the circular fence) will be represented by K:

$$K \rightarrow (x_K, y_K)$$

$$K \rightarrow ((1 - l),\ 0)$$

$$\therefore x_K = 1 - l$$

3.2. Defining the area

With the situation reformulated, integration techniques can be applied, in order to find the wanted area: T. Firstly, T will be divided into four parts:

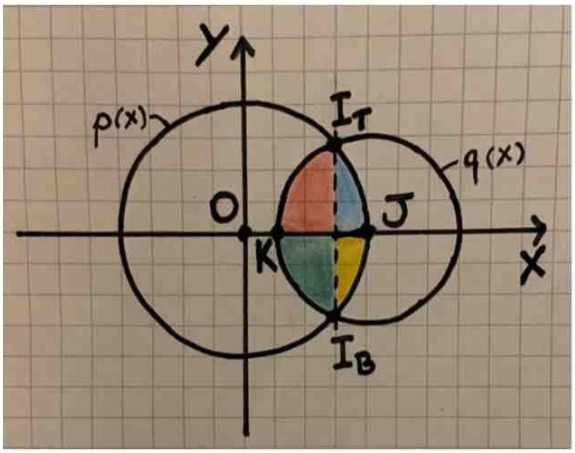


Figure 22: area T divided into four colored sections: red, blue, green, and yellow.

Thus:

$$T = S_{red} + S_{blue} + S_{green} + S_{yellow}$$

Since circles are symmetrical (and the horizontal axis of symmetry for both $c(x)$ and $d(x)$ is $y = 0$), the surface areas below the x-axis (S_{green} and S_{yellow}) are equal to the areas above it (S_{red} and S_{blue}):

$$S_{red} + S_{blue} = S_{green} + S_{yellow}$$

So T can be rewritten as:

$$T = 2(S_{red} + S_{blue})$$

Thereby, the whole analysis in this topic will be made considering only the top part of both circles in order to simplify things. Consequently, the interval for both functions is $y > 0$, meaning that:

- $p(x) = \ +\sqrt{1 - x^2}$
- $q(x) = \ +\sqrt{l^2 - (x - 1)^2}$

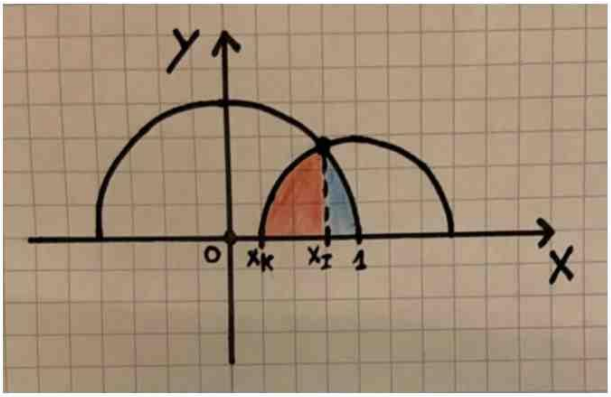


Figure 23: top part of functions $p(x)$ and $q(x)$, with x-values of intersections highlighted.

Finally, applying calculus to find the areas:

- $S_{red} = \int_{x_K}^{x_I} q(x)\, dx$
- $S_{blue} = \int_{x_I}^{1} p(x)\, dx$

Hence:

$$T = 2[\int_{x_K}^{x_I} q(x)\, dx + \int_{x_I}^{1} p(x)\, dx]$$

In the end, the function for *area T* should be in terms of θ. Therefore, a relation between x_I and x_K and θ should be established. As previously seen:

- $x_I \rightarrow$ x-coordinate of the point of intersection, I, between $p(x)$ and $q(x)$

$$p(x) = q(x)$$

$$\sqrt{1 - x^2} = \sqrt{l^2 - (x - 1)^2}$$

Squaring both sides:

$$1 - x^2 = l^2 - (x - 1)^2$$

$$1 - x^2 = l^2 - (x^2 - 2x + 1)$$

$$1 - x^2 = l^2 - x^2 + 2x - 1$$

$$1 = l^2 + 2x - 1$$

Isolating x:

$$2 - l^2 = 2x$$

$$x = \frac{2 - l^2}{2}$$

Using *relation 1*:

$$x = \frac{2 - (2cos(\frac{\theta}{2}))^2}{2}$$

$$x = \frac{2 - 4cos^2(\frac{\theta}{2})}{2}$$

$$x = 1 - 2cos^2(\frac{\theta}{2})$$

$$\therefore x_I = 1 - 2cos^2(\frac{\theta}{2})$$

- $x_K = 1 - l$

Using *relation 1*:

$$x_K = 1 - (2cos(\frac{\theta}{2}))$$

$$\therefore x_K = 1 - 2cos(\frac{\theta}{2})$$

This way, the definite integral above can be rewritten as:

$$T = 2[\int_{1-2cos(\frac{\theta}{2})}^{1-2cos^2(\frac{\theta}{2})} q(x)\,dx + \int_{1-2cos^2(\frac{\theta}{2})}^{1} p(x)\,dx]$$

3.3. Finding integrals

In order to simplify things, $p(x)$ and $q(x)$ will be integrated separately, before solving the huge definite integral found on the previous sub-topic.

❖ Starting with $p(x)$:

$$= \int p(x)\,dx$$

$$= \int \sqrt{1 - x^2}\,dx$$

Here, a trigonometric substitution can be applied ($x = sin(\alpha)$):

$$= \int \sqrt{1 - sin^2(\alpha)}\,dx$$

Applying the Pythagorean identity:

$$= \int cos(\alpha)\,dx$$

To continue integrating, both, the function being integrated "$cos(\alpha)$" and the differential of the variable x "dx", need to be in respect to the same variable. This way, the substitution previously stated will be used:

$$x = sin(\alpha)$$

Deriving both sides in respect to α:

$$\frac{d(x)}{d\alpha} = \frac{d(sin(\alpha))}{d\alpha}$$

$$\frac{dx}{d\alpha} = cos(\alpha)$$

$$\therefore dx = cos(\alpha) \cdot d\alpha$$

Substituting in the integral:

$$= \int cos(\alpha)\,(cos(\alpha) \cdot d\alpha)$$

$$= \int cos^2(\alpha)\, d\alpha$$

Applying the Double Angle Identity for cosine:

$$= \int \frac{cos(2\alpha)+1}{2}\, d\alpha$$

$$= \frac{1}{2}\int cos(2\alpha) + 1\, d\alpha$$

$$= \frac{1}{2}\,(\int cos(2\alpha)\, d\alpha + \int 1\, d\alpha)$$

To integrate "$cos(2\alpha)$" another substitution will be used:

$$u = 2\alpha$$

Deriving both sides in respect to α:

$$\frac{d(u)}{d\alpha} = \frac{d(2\alpha)}{d\alpha}$$

$$\frac{du}{d\alpha} = 2$$

$$\therefore d\alpha = \frac{1}{2}du$$

Using the substitution on the integration:

$$= \frac{1}{2}\,(\int cos(u)\,(\frac{1}{2}du) + \int 1\, d\alpha)$$

$$= \frac{1}{2}\,(\frac{1}{2}\,(\int cos(u)\, du) + \int 1\, d\alpha)$$

Finally, integrating:

$$= \frac{1}{2}\,(\frac{1}{2}\,(sin(u)) + \alpha)$$

$$= \frac{1}{2}\left(\frac{1}{2}(sin(2\alpha)) + \alpha\right)$$

$$= \frac{1}{4}(sin(2\alpha)) + \frac{\alpha}{2}$$

Yet, "p" is in terms of x. Consequently, the trigonometric substitution will be used once again:

$$x = sin(\alpha)$$

$$arcsin(x) = \alpha$$

So:

$$= \frac{1}{4}(sin(2arcsin(x))) + \frac{arcsin(x)}{2}$$

Lastly, a constant c has to be added since this equation is a result of an integration:

$$\int p(x)\,dx = \frac{1}{4}(sin(2arcsin(x))) + \frac{arcsin(x)}{2} + c$$

To simplify future calculations:

$$\int p(x)\,dx = P(x)$$

- Now, $q(x)$ will be integrated:

$$= \int q(x)\,dx$$

$$= \int \sqrt{l^2 - (x-1)^2}\,dx$$

A very important detail to remember is that, although it is being treated as a variable, l is a constant for integration purposes, since it represents a real value for the length of the rope. Thus, l can be isolated and thrown out of the integral:

$$= \int \sqrt{l^2\left(1 - \frac{(x-1)^2}{l^2}\right)}\,dx$$

$$= \int l \cdot \sqrt{1 - \left(\frac{x-1}{l}\right)^2}\,dx$$

$$= l\left(\int \sqrt{1 - \left(\frac{x-1}{l}\right)^2}\,dx\right)$$

Once again, a trigonometric substitution will be applied ($\frac{x-1}{l} = sin(\alpha)$):

$$= l(\int \sqrt{1 - sin^2(\alpha)}\, dx)$$

$$= l(\int cos(\alpha)\, dx)$$

As previously seen, both, the function being integrated "$cos(\alpha)$" and the differential of the variable x "dx", need to be in respect to the same variable. This way, the trigonometric substitution will be used:

$$\frac{x-1}{l} = sin(\alpha)$$

$$x - 1 = lsin(\alpha)$$

$$x = lsin(\alpha) + 1$$

Deriving both sides in respect to α:

$$dx = lcos(\alpha) \cdot d\alpha$$

Substituting in the integral:

$$= l(\int cos(\alpha)\, (lcos(\alpha) \cdot d\alpha))$$

$$= l^2(\int cos^2(\alpha)\, d\alpha)$$

It has already been shown how to integrate "$cos^2(\alpha)$". Thereby, some steps will be skipped:

$$= l^2(\frac{1}{4}(sin(2\alpha)) + \frac{\alpha}{2})$$

Naturally, this expression needs to be in terms of x. Therefore, the trigonometric substitution will be used one more time:

$$\frac{x-1}{l} = sin(\alpha)$$

$$arcsin(\frac{x-1}{l}) = \alpha$$

So:

$$= l^2(\frac{1}{4}(sin(2arcsin(\frac{x-1}{l}))) + \frac{arcsin(\frac{x-1}{l})}{2})$$

Lastly, using *relation 1* and adding the constant:

$$\int q(x)\,dx = (2cos(\tfrac{\theta}{2}))^2(\tfrac{1}{4}(sin(2arcsin(\tfrac{x-1}{2cos(\frac{\theta}{2})}))) + \frac{arcsin(\frac{x-1}{2cos(\frac{\theta}{2})})}{2}) + c$$

$$\int q(x)\,dx = (4cos^2(\tfrac{\theta}{2}))(\tfrac{1}{4}(sin(2arcsin(\tfrac{x-1}{2cos(\frac{\theta}{2})}))) + \frac{arcsin(\frac{x-1}{2cos(\frac{\theta}{2})})}{2}) + c$$

To simplify future calculations:

$$\int q(x)\,dx = Q(x)$$

3.4. Determining the 3rd Function

Now, to finally find the 3rd function, the definite integral found in topic 3.2 will be solved by using the information found in topic 3.3:

$$T = 2[\int_{1-2cos(\frac{\theta}{2})}^{1-2cos^2(\frac{\theta}{2})} q(x)\,dx + \int_{1-2cos^2(\frac{\theta}{2})}^{1} p(x)\,dx]$$

$$T = 2[Q(1 - 2cos^2(\tfrac{\theta}{2})) - Q(1 - 2cos(\tfrac{\theta}{2})) + P(1) - P(1 - 2cos^2(\tfrac{\theta}{2}))]$$

By solving this, an expression with only one variable (θ, in the case) would be found. However, since the integrals of p(x) and q(x), P(x) and Q(x) respectively, are huge equations, this 3rd function for T (expanded) wouldn't fit the page. This way, the function will be left as it is above and, thereby, the 3rd function for *area T* in terms of θ has been found.

Plotting this 3rd function on a Cartesian Plane:

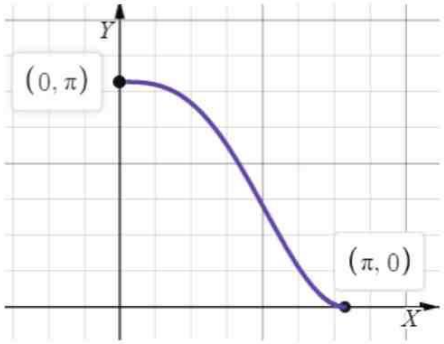


Figure 24: 3rd function for area T plotted on a Cartesian plane.

4. Conclusion

In this work, two different approaches to the same problem were presented. In the end, the goal was achieved – three different representations for *area T* were found. Here are the resulting functions:

1) $T(\theta) = \theta cos(\theta) + \pi - sin(\theta)$

2) $T(\theta) = \pi - \frac{1}{3}\theta^3 + \frac{1}{30}\theta^5 - \frac{1}{840}\theta^7 + \frac{1}{45360}\theta^9$

3) $T(\theta) = 2[Q(1 - 2cos^2(\frac{\theta}{2})) - Q(1 - 2cos(\frac{\theta}{2})) + P(1) - P(1 - 2cos^2(\frac{\theta}{2}))]$

And here are the graphs:

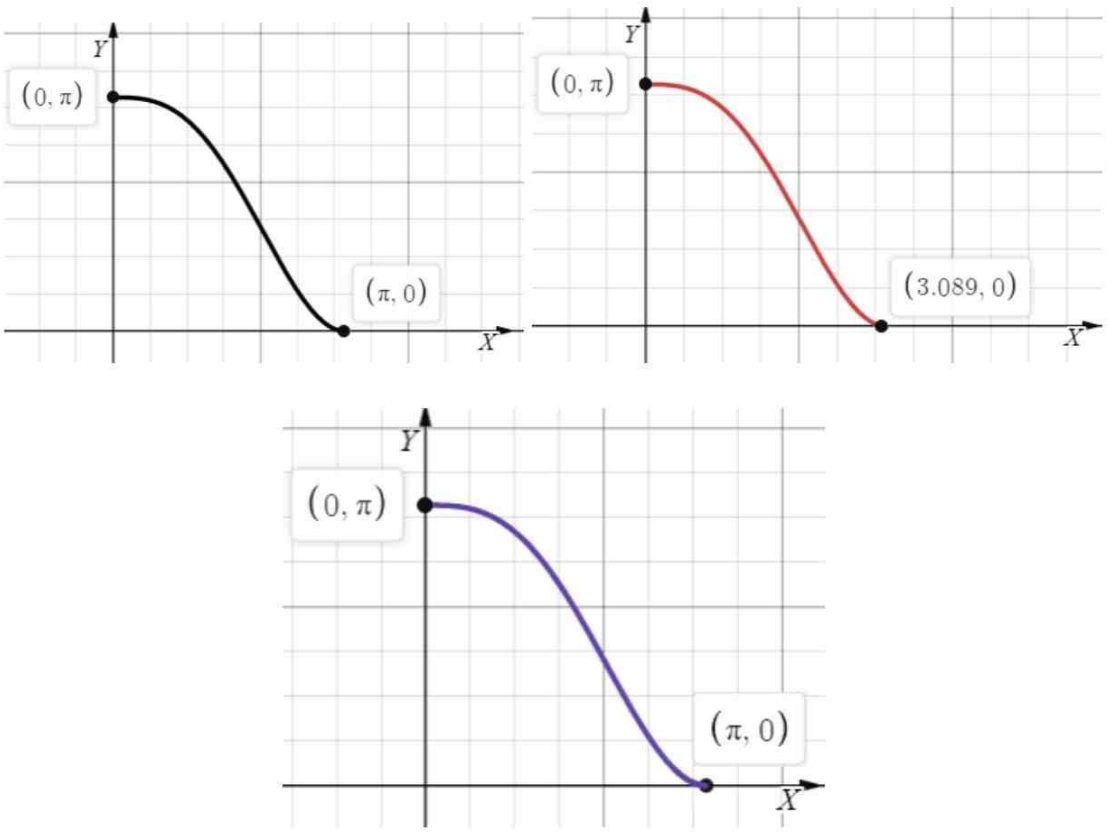


Figure 25: Graphs for functions 1) (in black), 2) (in red) and 3) (in purple).

After working out all the mathematics in this work, it is clear that the trigonometric approach to this problem is way more practical and it gives a way more pleasing-to-the-eye

solution. Nevertheless, approaching it through algebra/calculus is really interesting and fun, because it presents the same problem through a whole different perspective and makes the one trying to solve it “think out of the box”. Furthermore, as previously mentioned, finding the same answer to the same problem through different approaches increases the credibility of the solution.

Lastly, a suggestion for further research into this problem could be trying to solve it through another, different approach, using other areas inside mathematics – is it even possible? Furthermore, this same problem could be discussed for a three-dimensional situation: a spherical cage of a bird, for example.

5. References

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